GATE-AIII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 03



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Recap of Previous Lecture



Topic Existence of limits

Topics to be Covered



Topic	How to evaluate the limits?



Topic : Single Variable Calculus

#Q. If $f(x)=\begin{cases}\dfrac{x-|x|}{x} & , \quad x\neq 0\\ 2 & , \quad x=0\end{cases}$ the $\lim\limits_{x\to 0} f(x)$ is

A 2

B 0

C 1

D Does not exist

$$f(x) = \begin{cases} \frac{x - [x]}{x} & x \neq 0 \\ 2 & x = 0 \end{cases}$$

$$RHL = \lim_{h \to 0} f(0+h) = \lim_{h \to 0} f(h)$$

$$= \lim_{h \to 0} \frac{h - |h|}{h} = 0$$

$$LHL \neq RHL$$

Left Hand limit

$$LHL = \lim_{h \to 0} f(a-h) \quad a = 0$$

$$= \lim_{h \to 0} f(-h)$$

$$= \lim_{h \to 0} \frac{-h - |-h|}{-h}$$ (positive h)

$$= \lim_{h \to 0} \frac{-h-h}{-h} = \lim_{h \to 0} \frac{-2h}{-h}$$

Limit does Not exists $= 2$



Topic : Single Variable Calculus

Indeterminate forms

Examples : $x \to$ approaches

$$\left\{ \frac{\to 0}{\to 0} \right. , \quad \frac{\to \infty}{\to \infty} , \quad (\to \infty) - (\to \infty)$$

$$\to \infty \times \to 0 , \quad (\to 1)^{\to \infty} , \quad (\to 0)^{\to 0} , \quad (\to 1)^{\to \infty}$$

Total 7 Indeterminate forms

Indeterminate forms

$$\frac{f(x)}{g(x)} = \frac{\to \infty}{\to \infty} , \frac{\to 0}{\to 0}$$

$$= (\to 1)^{\to \infty}$$

$$(\to \infty)^{\to \infty}$$



Topic : Single Variable Calculus

$\frac{\to 0}{\to 0}, \frac{\to \infty}{\to \infty}, \to \infty - \to \infty$

$(\to \infty)^{\to \infty}, (\to 1)^{\to \infty}$

✓ Strategy
✓ Indeterminate Form – remove
✓ Plug-in Limits

#Q. Evaluate: $\lim_{x \to 1} \left(4x^3 - 3x^2 + 6\right)$.

$= \lim_{x \to 1} (4x^3 - 3x^2 + 6)$

Free for all Indeterminate form

↓ Limit Plug-in

$= \lim_{x \to 1} (4x^3 - 3x^2 + 6)$

$= 4(1)^3 - 3 \times (1)^2 + 6$

$= 7$



Topic : Single Variable Calculus

#Q. Evaluate: $\lim_{x \to 0} \dfrac{\cos^3 x - 3\cos x + 7}{3x^2 + 5x - 14}$

$\left\{\frac{\to 0}{\to 0}\right\}$

Check It

Free form ⟶ Indeterminate form

Direct Plug-in Rule.

$= \dfrac{1 - 3 + 7}{0 + 0 - 14} = -\dfrac{5}{14}$ Ans

Topic : Single Variable Calculus

(i) Evaluate: $\lim_{x\to 2} \dfrac{x^3-2x-4}{x^2-3x+2}$

$\dfrac{(2)^3-2\times2-4}{(2)^2-3\times2+2} = \dfrac{\to 0}{\to 0}$ ↓ go to 8th

Indeterminate form

1st strategy Remove The Indeterminate Form

Tool (Using Factorization)

Using Factorization

$\lim_{x\to 2} \dfrac{(x-2)(x^2+2x+2)}{(x-1)(x-2)}$

$= \lim_{x\to 2} \dfrac{x^2+2x+2}{(x-1)}$ remove It

↓ Plug in Limit

$= \dfrac{(2)^2+2\times2+2}{2-1}$

$= \dfrac{4+6}{1} = 10$ Ans

$x^2-2x-x+2$

$x(x-2)-1(x-2)$

$(x-1)(x-2)$



Topic : Single Variable Calculus

(ii) Evaluate: $\lim_{x \to a} \frac{x^3 - a^3}{x^2 - ax}$

$= \lim_{x \to a} \frac{x^3 - a^3}{x^2 - ax} = \frac{\to 0}{\to 0}$ Indeterminate Forms

STRATEGY No-1

Using factorization

$= \lim_{x \to a} \frac{(x-a)(x^2+a^2+ax)}{x(x-a)}$ Remove lt

$= \lim_{x \to a} \frac{x^2+a^2+ax}{x}$

$=$ Plug in Limits

Plug in

$= \frac{a^2+a^2+a^2}{a}$

$= \frac{3a^2}{a} = 3a$ Ans



Topic : Single Variable Calculus

(iii) Evaluate: $\lim_{x\to 2} \dfrac{x^3-6x^2+11x-6}{x^2-6x+8}$

factor

$\dfrac{\to 0}{\to 0}$] Indeterminate form

remove It

Plug-in Limits

Using Factorization

$$= \lim_{x\to 2} \frac{(x-1)(x-2)(x-3)}{(x-2)(x-4)}$$

$$= \frac{(2-1)(2-3)}{(2-4)}$$

Remove It + Plug-in

$$= \frac{1\times(-1)}{-2} = \frac{1}{2}$$ Ans



Topic : Single Variable Calculus

#Q. Evaluate: $\lim_{x \to 3} \frac{x^3 - 7x^2 + 15x - 9}{x^4 - 5x^3 + 27x - 27}$

A 2/9

B 1/9

C -2/9

D None of these



Topic : Single Variable Calculus

#Q. Evaluate: $\lim_{x\to 5} \dfrac{x^4-625}{x^3-125}$ $\Rightarrow \dfrac{\to 0}{\to 0}$ form Indeterminate form

Strategy – Using factorization

$$\lim_{x\to 5} \frac{x^4-(5)^4}{(x-5)}$$ Temp ① divide via $(x-5)$

$$\frac{x^3-(5)^3}{(x-5)}$$ Temp ②

$$\Rightarrow \frac{4\times(5)^{4-1}}{3\times(5)^{3-1}} = \frac{4\times 5^3}{3\times 5^2} = \frac{20}{3}$$ Ans.

$$\lim_{x\to a} \frac{x^n-a^n}{(x-a)} = \text{Template wallpaper}$$

$$\lim_{x\to a} \frac{x^n-a^n}{(x-a)} = na^{n-1}$$



Topic : Single Variable Calculus

#Q. Evaluate: $\lim_{x\to 3} \dfrac{\sqrt{3x+7}-4}{\sqrt{x+1}-2}$

$\sqrt{a}+b = \sqrt{a}-b$ → Rationalize

$= \dfrac{\sqrt{16}-4}{2-2}$

$= \dfrac{\to 0}{\to 0}$ "Indeterminate Form"

✕ ST rategy ① Using factori
② Template I

Strategy No-2 → Using Rationalization

$$\lim_{x\to 3} \frac{\sqrt{3x+7}-4}{\sqrt{x+1}-2} \times \frac{\sqrt{x+1}+2}{\sqrt{x+1}+2} \times \frac{\sqrt{3x+7}+4}{\sqrt{3x+7}+4}$$

$$= \lim_{x\to 3} \frac{(3x+7)-16}{(x+1)-4} \times \frac{\sqrt{x+1}+2}{\sqrt{3x+7}+4}$$

$$= \lim_{x\to 3} \frac{3(x-3)}{(x-3)} \times \frac{\sqrt{x+1}+2}{\sqrt{3x+7}+4}$$

$$= 3 \times \frac{(2+2)}{(4+4)} = \frac{12}{8} = \frac{3}{2}$$

$= 1.5$



Topic : Single Variable Calculus

#Q. Evaluate: $\lim\limits_{x\to a}\dfrac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}} = \dfrac{\to 0}{\to 0}$ Indeterminate form

Using Rationalize = strategy

$$= \lim_{x\to a}\frac{(\sqrt{a+2x}-\sqrt{3x})}{(\sqrt{3a+x}-2\sqrt{x})}\times\frac{(\sqrt{a+2x}+\sqrt{3x})}{(\sqrt{a+2x}+\sqrt{3x})}\times\frac{\sqrt{3a+x}+2\sqrt{x}}{(\sqrt{3a+x}+2\sqrt{x})}$$

$$= \lim_{x\to a}\frac{(a+2x-3x)}{(3a+x-4x)}\,\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{a+2x}+\sqrt{3x}}$$

$$= \frac{1}{3}\lim_{x\to a}\frac{\sqrt{3a+x}+2\sqrt{x}}{\sqrt{a+2x}+\sqrt{3x}} = \frac{2}{3\sqrt{3}}$$



Topic : Single Variable Calculus

#Q. Evaluate: $\lim_{x\to\infty}\left(x-\sqrt{x^2+x}\right)$

Solve It

$\Rightarrow \to\infty - \to\infty$

Indeterminate

Strategy 02

remove The Indeterminate

Using Rationalization

A	-1/2	B	1/2
C	1	D	-1

$$\lim_{x\to\infty}\left[x-\sqrt{x^2+x}\right]$$

$$=\lim_{x\to\infty}\left[x-\sqrt{x^2+x}\right]\frac{x+\sqrt{x^2+x}}{x+\sqrt{x^2+x}}$$

$$=\lim_{x\to\infty}\left[\frac{x^2-(x^2+x)}{x+\sqrt{x^2+x}}\right]$$

$$=\lim_{x\to\infty}\left[\frac{-x}{x+\sqrt{x^2+x}}\right]$$

$$=\lim_{x\to\infty}\left[\frac{-1}{1+\sqrt{\frac{x^2}{x^2}+\frac{x}{x^2}}}\right]$$ divide x

"Two Important Rule" $\left\{\begin{array}{l} x\to\infty \\ \frac{1}{x}\to 0 \end{array}\right.$

Wrong concept

$$=\lim_{x\to\infty}\left[\frac{-1}{1+\sqrt{1+\frac{1}{x}}}\right]$$

Plug in Limit

$$=\frac{-1}{1+\sqrt{1+0}}$$

$$=\frac{-1}{1+1}=\boxed{\frac{-1}{2}}$$



Topic : Single Variable Calculus

#Q. Evaluate: $\lim\limits_{x\to\infty} \dfrac{x^3-2x^2+3x+1}{5x^3+7x+2}$

$\Rightarrow \dfrac{\to\infty}{\to\infty}$ Indeterminate form

Strategy $x\to\infty$, $\frac{1}{x}\to 0$

$$= \lim_{x\to\infty} \frac{1-\frac{2}{x}+\frac{3}{x^2}+\frac{1}{x^3}}{5+\frac{7}{x^2}+\frac{2}{x^3}}$$

$$= \frac{1}{5} \text{ Ans}$$



Topic : Single Variable Calculus

#Q. Evaluate: $\lim_{n\to\infty} \dfrac{1^2+2^2+\ldots\ldots\ldots+n^2}{n^3}$

$$\lim_{n\to\infty}\left[\frac{1^2}{n^3}+\frac{2^2}{n^3}+\cdots+\frac{n^2}{n^3}\right]$$

① ② $+n$

Infinite Pieces.

$$1^2+2^2+3^2+\cdots+n^2 = \sum n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\frac{(n^2+n)(2n+1)}{2n^3+3n^2+n} = \lim_{n\to\infty} \frac{1^2+2^2+\cdots+n^2}{n^3}$$

$$= \lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^3}$$

$$= \lim_{n\to\infty} \frac{2n^3+3n^2+n}{6n^3}$$

$$= \lim_{n\to\infty} \frac{2+\frac{3}{n}+\frac{1}{n^2}}{6} \quad \left(\frac{3}{n}\to 0,\ \frac{1}{n^2}\to 0\right) = \frac{2}{6} = \frac{1}{3}$$

Using strategy $\left[n\to\infty,\ \frac{1}{\infty}\to 0\right]$



Topic : Single Variable Calculus

#Q. The value of $\lim_{x\to 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

$\frac{1-2+1}{1-3+2} = \frac{\to 0}{\to 0}$ form

Using factorization

$x^5 = t$

(A) is 0

(B) is – 1

(C) is 1

(D) Does not exit

→ NEXT – L-Hospital Rule

$\lim_{x\to 0}$ = method work

$\lim_{x\to 1}$ = L-Hospital Rule



Topic : Single Variable Calculus

#Q. Compute $\lim_{x\to 3} \frac{x^4-81}{2x^2-5x-3}$

A 1

B limit does not exit

C 53/12

D 108/7

$2x^2-6x+x-3$

$2x(x-3)-1(x-3)$

$(2x-1)(x-3)$

Using factorization

$$\frac{x^2-(9)^2}{2x^2-5x-3}$$

$$= \frac{(x^2-9)(x^2+9)}{(2x-1)(x-3)}$$

$$\text{lt}_{x\to 3} \frac{(x-3)(x+3)(x^2+9)}{(2x-1)(x-3)} = \frac{108}{7}$$

Differentiation
12th

THANK - YOU